Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CS54]

# FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

## SOFTWARE ENGINEERING & TESTING

**Time: 3 Hours** **Maximum Marks: 70**

**Instructions:**
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

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| **1.** | **a.** | What is software engineering? List and explain the activities of software process | **07** |
| | **b.** | List the different professional and ethical responsibilities to be followed by any software engineer. | **07** |
| **2.** | **a.** | List the different users of requirement document. Illustrate the IEEE standard to explain the structure for requirement document. | **07** |
| | **b.** | Bring out the need of requirement elicitation and analysis process. Explain it with a neat diagram. | **07** |
| **3.** | **a.** | Write neatly Use case diagram, UML Sequence diagram and UML Collaboration diagram with briefly explained notations for Telephone call. | **07** |
| | **b.** | Draw UML diagram for the following with suitable example:<br>**i.** Association for a student who works as a intern and is associated as a trainer also.<br>**ii.** Composition<br>**iii.** Component and deployment diagram for client server architecture cconstraint | **07** |

**OR**

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| **4.** | **a.** | Highlight the significance of Component diagrams and Deployment diagrams. | **07** |
| | **b.** | Write the UML class diagrams and Use case diagrams for Bill processing system of purchasing any electronic goods. | **07** |
| **5.** | **a.** | What are Architectural Patters? Explain MVC Pattern in Detail. | **07** |
| | **b.** | Define design Principles. Explain the four essential elements of pattern. | **07** |

**OR**

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| **6.** | **a.** | Explain how design pattern helps in specifying object implementation. Write Advantage and disadvantage of Delegation. | **07** |
| | **b.** | Describe common causes of redesign along with the patternsto address them. | **07** |
| **7.** | **a.** | Explain Integration testing strategies. Address the issues related to integration testing | **07** |
| | **b.** | Briefly describe about<br>**i.** System Testing with examples.<br>**ii.** Acceptance testing with examples. | **07** |

**OR**

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| **8.** | **a.** | Explain Functional and Structural Testing. Also explain how it can impact software testing. | **07** |
| | **b.** | Explain Metric Based Testing in software testing. Explain with an example. | **07** |